ViCivil-TKG: A Temporally Anchored Benchmark for Vietnamese Civil-Law Question Answering with Multi-Round Gold-Answer Adjudication

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Open project ViCivil-TKG — https://huggingface.co/datasets/datbazuka2502/vicivil-tkg · https://github.com/ntd25022006q/vicivil-tkg

Abstract. We introduce **ViCivil-TKG**, a benchmark for Vietnamese civil-law question answering built around two under-addressed challenges: (i) *temporal validity* — the same question has different answers under different applicability windows of a statute; and (ii) *gold-label reliability* — most automatically generated questions carry unverified gold answers. The benchmark publishes **1,092 validated questions** (a 281-question core with 12 hand-written and 269 controlled-generated items; 106 round-1 extension items; 64 round-2 items from the 1996–2015 statute era; 265 round-3 and 376 round-4 scale-up items) over a corpus of 20 legal documents with 13,151 passages, together with **1,029 publicly quarantined questions** that count in no metric. Every gold answer passes a four-stage adjudication chain: two-tier generation-time auditing, two-round re-anchoring with a larger judge, a pass-3 sweep over 99 weak-gold suspects with three guardrails, and an agreement audit on a fixed 60-question sample; afterwards no question remains unverifiable. On the core set, temporal-consistency retrieval (TCR) raises the citation accuracy of Qwen2.5-3B from 0.488 to 0.601 (McNemar $p = 5.8 \times 10^{-5}$; 0.004 without context) and retrieval reaches Hit@5 0.811; on the 1996–2015 cohort TCR lifts Hit@5 from 0.703 to 0.781 and reranking raises Hit@1 to 0.703; on the 641-question round-3/4 scale-up TCR raises Hit@1 0.511→0.654 (round 4, $p = 6.3 \times 10^{-22}$) and the same chain lifts cite_acc 0.521→0.600 (round 3, $p = 3.8 \times 10^{-3}$) and 0.527→0.657 (round 4, $p = 3.2 \times 10^{-10}$). Full adjudication trails, a logged data-integrity audit, and a GPU-free reproduction script are released.

Keywords: legal question answering; Vietnamese datasets; temporal retrieval; gold-label adjudication; RAG.

1 Introduction

Legal question-answering systems must be considerably more accurate than open-domain ones: citing the wrong article has real legal consequences. Building an evaluation suite for this domain runs into two obstacles. First, Vietnamese statutes have explicit *effective dates*: the 2015 Civil Code replaced the 2005 Code, ministerial decrees are replaced on a rolling cycle, so the question "how long is the limitation period for a claim to recover property?" must be answered from the statute that was in force *at the question's time point*, not from the newest one. Second, QA sets generated automatically with LLMs assume the gold answer is correct, whereas in practice gold anchoring — pointing to the exact passage of law that grounds the answer — is the largest source of error.

This paper presents ViCivil-TKG, a benchmark covering both aspects. The contributions are: **(C1)** a benchmark of 1,092 published questions over a corpus of 20 statutes (13,151 passages), each question carrying an `as_of` date and a gold anchor to an exact passage identifier, plus 1,029 publicly quarantined questions; **(C2)** a *four-stage gold adjudication chain* with multiple independent judge models, guardrails on every application, full provenance, and public quarantine of rejected items; **(C3)** a temporal-consistency retrieval (TCR) protocol with paired statistical tests for both retrieval and generation; and **(C4)** a *data-integrity audit* of the entire release in which every discrepancy found was fixed with a logged trail — a practice we argue should be standard for benchmark datasets.

2 Related Work

Retrieval and QA benchmarks. BEIR [4] standardises zero-shot retrieval evaluation across domains; LexGLUE [5] aggregates English legal NLP tasks. Neither provides temporal-validity segmentation nor Vietnamese statutes, and their gold labels are expensively human-verified — unlike our controlled generation with multi-round adjudication. SQuAD 2.0 [6] introduced unanswerable questions, close in spirit to the abstention behaviour we measure through the rate of "insufficient information" replies.

Dense retrieval and fusion. BM25 [1] remains a strong retrieval backbone; DPR [3] adds a semantic signal; Reciprocal Rank Fusion [2] merges two rankings without training. We use exactly BM25 + dense + RRF in every experiment, and add the temporal-validity filter (TCR) built on the statute timeline graph. BGE-reranker [8] serves as the reranking stage; RAG [7] and the Qwen family [9] are the substrate for the cited-generation conditions.

Temporal and legal QA benchmarks. Temporal QA over encyclopaedic sources is served by TempQuestions [10] (1,271 questions over Freebase), TimeQA [11] (41.2k time-sensitive questions), SituatedQA [12] (12.2k questions under temporal/geographic context shifts), and, at much larger scale, ComplexTempQA [13]; Su et al. [14] survey temporal KG-QA. In all of these, time is an attribute of the *asked-about fact*; the answer source itself has no validity window. On the Vietnamese legal side, the Zalo legal-retrieval shared tasks [16] (3,196 questions over 61.4k articles in the 2022 edition), VLQA [17] (∼3.1k annotated questions), ViLegalExpert [18] (over 172k citizen–lawyer consultation pairs), and VLegal-Bench [19] (10,450 expert-labelled samples) cover statutes well but carry no temporal anchoring, and French

LLeQA [15] (1,868 expert-annotated legal questions) neither. Table 1 positions ViCivil-TKG: to the best of our knowledge it is the first benchmark that combines (i) a statutory domain, (ii) per-question `as_of` anchoring scored against exact passage identifiers, and (iii) a multi-round gold adjudication trail with public quarantine.

3 The ViCivil-TKG Dataset

3.1 Corpus of legal documents

The corpus covers 20 Vietnamese civil-law documents from 1995–2026 (the Civil Codes of 1995/2005/2015, the 2004 Civil Procedure Code and its consolidated text, the Marriage and Family Laws of 2000/2014, the 2008 Civil Judgment Enforcement Law and its consolidation, the 2010 Adoption Law, the State Compensation Liability Law, implementing decrees, and newly enacted 2026 documents), segmented into **13,151 passages** along the article–clause–point structure. Every passage carries mandatory provenance: `source_url`, `source_type` (official / official-copy / semi-official / mirror), `fetched`, and the validity window `valid_from`/`valid_to` taken from a temporal graph of 37 documents with 24 succession–amendment edges. Two passages extracted from a corrupted `.doc` conversion were dropped and logged; passage counts were re-counted directly on the released copy during the release-integrity audit.

3.2 Controlled question generation

Questions are generated locally with `qwen2.5:14b`, batch-wise from source passages, under three constraints: (i) the `as_of` date must lie inside the validity window of the source statute; (ii) the question must be answerable *only* from the source passage and its directly related articles; (iii) within-batch deduplication uses question vectors (768-dim, `embeddinggemma:300m`). The core set contains 269 generated plus 12 hand-written questions (CIV001–CIV012, one of which probes the abstention behaviour when the governing statute is not yet in force). Two extension waves added 106 published questions (round 1, from 242 generated) and 154 questions generated from three 1996–2015-era documents (the 2004 Civil Procedure Code, Decree 158/2005 on civil status, and Decree 126/2014 implementing the Marriage and Family Law) with quotas 103/30/27. A third scale-up wave (round 3) sampled 620 passages under per-law quotas (150 per statute), generated 590 valid questions, dropped 9 near-duplicates (token-set Jaccard ≥ 0.60 against all prior questions), and published 265 after the same two-tier audit and re-anchoring; 316 were quarantined. A fourth scale-up wave (round 4) repeated the recipe against the enlarged corpus snapshot: 938 questions were generated, 75 embedding near-duplicates of already-published questions were dropped before auditing (comparison log `extension4_dedup.json`), the remaining 863 were audited under the same two-tier chain (accounting $863 = 376 + 487$, exact), 13 golds were re-anchored after two-round confirmation, and 376 were published (363 verified + 13 regolded) while 487 were quarantined; every published question carries a unique `qid` and 0 `as_of` repairs were needed. A release audit caught 28 question-ID collisions produced by parallel generation (two different questions sharing one ID, making their stored verdicts ambiguous); every affected pair was re-adjudicated from scratch under unique IDs and the repair log is released (`runs/extension3/qid_repair_log.json`). 28 of the 281 core questions originally carried an `as_of` date before the effective date of their gold statute; all were moved into the correct validity phase with a full log (`as_of_fixes.json`). Extension-cohort questions derive `as_of` from the graph at generation time (0 repairs).

3.3 The gold adjudication chain

This is the central contribution. The four-stage chain (Figure 1) leaves its full trail in `benchmark/`:

1. **Two-tier audit at generation time.** Every generated question is scored immediately by an 8B judge (`usable: yes/partial/no`); items that fail "yes" are re-scored by a 14B judge with an evidence-quote requirement. Items failing both tiers are publicly quarantined.

2. **Two-round re-anchoring.** For questions with suspect golds, the regold pipeline proposes a replacement passage and asks a supervising LLM to confirm it in *two independent rounds*; only passages accepted by both rounds are applied. On the core set, 38 questions were re-anchored this way (two-round agreement 0.888).

3. **Pass-3: a sweep over 99 weak-gold suspects.** The entire weak-gold group (never-verified, re-golded, multi-part, and hand-written items) is re-scored by a deeper judge (`qwen3:14b`) under the protocol "current gold + 8 stored retrieval candidates": verdicts are *gold-ok* (66), *reanchor* (3, plus 1 blocked by a guardrail because the proposal was not among the stored candidates), or *flag* (29). Every application must pass three guardrails: the passage must exist in the corpus, must appear in the stored candidate list, and must be valid under the temporal window at `as_of`. The final 2 "unverifiable" questions were rechecked separately and both were successfully re-anchored to concrete passages.

4. **Independent agreement audit.** On a fixed 60-question sample (seed 20261101), an independent judge chooses the legal basis among 8 candidates: 56/60 parse successfully, and 31/56 pick the applied gold set (**0.554**), with 32/56 echoing the primary gold passage (**0.571**). We publish these numbers *unfiltered* as a measure of how hard gold anchoring is in this domain, and stress that this is a model-vs-model check, not a substitute for human evaluation (Section 6); the 60-question human adjudication kit with a Cohen/Fleiss κ script is released under `docs/annotation/` as three randomized per-annotator packets (fixed seeds; each packet exposes only its own judgment column, enforcing independent scoring), together with a clearly-labelled synthetic fixture on which the κ implementation reproduces independent reference values exactly; the human run itself is the required next step (Section 6).

After the chain, **281/281 core questions carry a verified gold**. The final tier structure is 229 `verified`, 22

Table 1. Positioning against temporal-QA and Vietnamese-legal benchmarks. "Time anchoring" means the question fixes the validity time point of the *answer source*, not of the asked-about fact. Sizes are quoted from the respective papers.

Benchmark	Domain	Lang.	Size	Time anchoring	Gold protocol
TempQuestions [10]	open KB (Freebase)	EN	1,271 q	temporal fact constraints	crowdsourced
TimeQA [11]	Wikipedia	EN	41.2k q	time-sensitive facts	crowdsourced
SituatedQA [12]	Wikipedia	EN	12.2k q	context time/place shift	crowdsourced
ComplexTempQA [13]	Wikipedia/Wikidata	EN	100M pairs	time-scope metadata	auto + filters
LLeQA [15]	statutory law	FR	1,868 q	none	expert-annotated
Zalo Legal retrieval [16]	VN statutes (retrieval)	VN	3,196 q / 61.4k articles	none	shared task
VLQA [17]	VN law	VN	∼3.1k q	none	human-verified
ViLegalExpert [18]	VN law (retrieval)	VN	172k q	none	mined consultations
VLegal-Bench [19]	VN law (multi-task)	VN	10,450 items	none	expert cross-validation
ViCivil-TKG (ours)	VN civil statutes	VN	**1,092 q (+1,029 quarantined)**	**per-question `as_of` validity windows**	multi-round adjudicated + quarantine

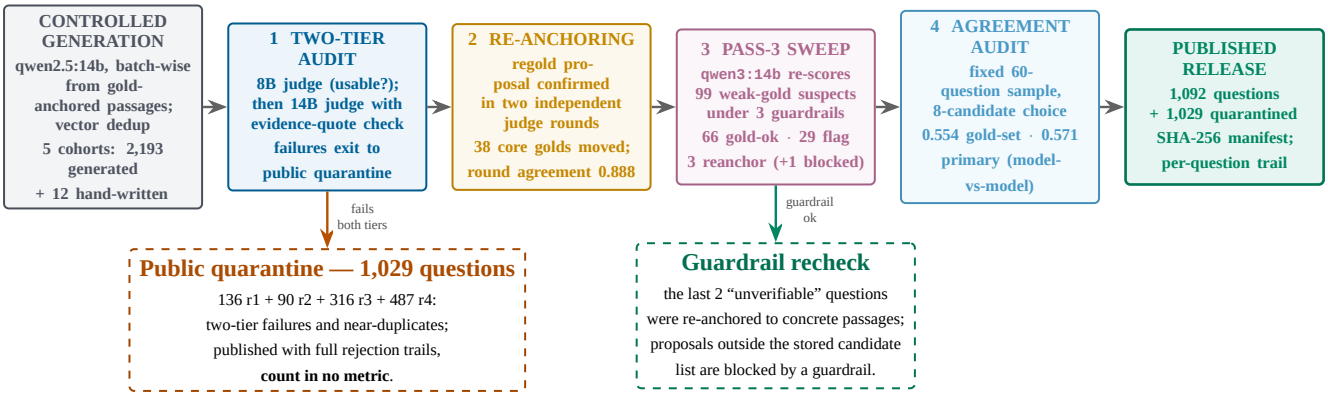


Figure 1. The four-stage gold adjudication chain. Controlled generation (Section 3.2) feeds stage 1; every branch leaves a public trail, and quarantined questions never enter any metric.

`regolded`, 4 `author-verified`, 26 `flagged` — in the release-integrity audit (Section 5.7) we found that the pass-3 reanchor of one question (TOP27031) had been recorded in the adjudication file but never persisted into the question file, so scoring used the stale gold (rank 31 instead of rank 1); the fix was applied with a log entry, the tier moved from `flagged` to `regolded`, and all metrics were recomputed. 26 questions carry the `flagged` tier and are excluded from all primary metrics, with reasons published.

3.4 Cross-cohort deduplication

Round 2 adds the step that round 1 lacked: cosine similarity between every newly generated question and all 387 already-published questions (threshold 0.95). One near-exact duplicate of a core question (cosine 0.9931) was quarantined; 0 questions matched the round-1 quarantine group. The comparison log is stored in `extension2_dedup_audit.json`. Hence the 154 round-2 generated questions yield 64 published + 90 quarantined (89 two-tier failures + 1 duplicate). Rounds 3 and 4 repeat the scan against *all* published questions: 9 near-duplicates were dropped in round 3 (token-set Jaccard) and 75 in round 4 (embedding cosine; each entry in `extension4_dedup.json` lists the dropped question and its nearest published neighbour).

3.5 Release structure

The benchmark releases **1,092 questions** over five cohorts (281 core, 106 round-1, 64 round-2, 265 round-3, 376 round-4) and **1,029 quarantined questions** (136 round-1 + 90 round-2 + 316 round-3 + 487 round-4), each with a complete per-question trail. The core set is split stratified dev/test 140/141. Every extension cohort publishes stratified metrics by gold tier; quarantined questions *never* enter any metric. All artifacts carry SHA-256 hashes and passed a 12-point audit (manifest, JSON parse-all, matrix shape, gold chains, extraction coverage, secret scan, schema, duplicate accounting, validity-window sanity, kv references, odd-character gate, identifier set).

4 Experimental Setup

Retrieval. Unified protocol: BM25 (pure Python, k_1=1.2, b=0.75) + 768-dim dense vectors (stored passage vectors, `embeddinggemma:300m`) fused by RRF k=60, drawing 200 candidates per branch. Scoring is strict gold-passage-identifier matching; we report Hit@{1,3,5,10} and MRR. *MRR semantics:* the core TCR results use full-list MRR; the no-TCR repro is capped at the top-10 window (MRR@10) because of its re-scoring protocol; the gold-effect tables use MRR within the stored 30-candidate window (MRR@30) — each is annotated separately to avoid cross-protocol comparisons. TCR (temporal-consistency retrieval) filters candidates by the validity window of their statute at the question's `as_of`; statutes outside the graph pass through unchanged. For extension cohorts a BGE-reranker-v2-m3 stage over 30 candidates is applied before TCR. Statistical significance: paired Wilcoxon on the reciprocal rank of the first gold hit.

Cited generation. Qwen2.5-3B-Instruct (greedy, max 180 new tokens, batch 6) runs under four conditions: *norag* (no context), *rag_hybrid* (top-5 fused), *rag_hybrid_tcr* (top-5 af-

Table 2. Retrieval on the 281-question core (hybrid BM25 + dense + RRF, temporal-consistency filter TCR; full-list MRR).

Stratum	n	Hit@1	Hit@3	Hit@5	Hit@10	MRR
All	281	**0.541**	**0.740**	**0.811**	**0.879**	**0.662**
verified	229	0.585	0.760	0.821	0.895	0.693
regolded	22	0.273	0.773	0.864	0.909	0.532
auth.-verified	4	0.250	0.500	0.750	0.750	0.403
flagged (out of metrics)	26	0.423	0.577	0.692	0.731	0.538
Repro without TCR (MRR@10)	281	0.399	0.651	0.733	0.815	0.540

ter TCR), *rag_rerank_tcr* (top-5 after rerank + TCR). We measure *cite_acc* (the answer names the correct article numbers of the gold set), *span_F1* (token F1 against the gold text), *abstain* (rate of "insufficient information" replies), and *ctx_hit5* (the gold set appears in the context). Significance: exact McNemar for cite_acc, Wilcoxon for span_F1.

Reproduction. All retrieval metrics and the LLM rescore are computed from stored artifacts (vectors, rankings, answers) and re-runnable with `tools/reproduce.sh` without GPU or network.

5 Results

5.1 Retrieval on the core set

Table 2 reports the 281 core questions under the documented hybrid+TCR protocol: Hit@5 **0.811** and MRR **0.662** overall; the `verified` stratum (229 questions) reaches Hit@5 0.821 and MRR 0.693. The three weaker strata are published verbatim: `flagged` (26 questions, Hit@5 0.692), `regolded` (22 questions, Hit@1 0.273 but Hit@5 0.864), and `author-verified` (4 questions). The "low Hit@1, high Hit@5" pattern of the regolded stratum shows that reanchored passages often sit deep in the candidate list — exactly the case that multi-round adjudication prevents from being scored as a miss. The no-TCR repro (same scoring code, no temporal filtering) reaches Hit@5 0.733 vs. 0.811: the gap is the contribution of temporal filtering, and the gain reproduces on all four extension cohorts (Figure 2).

5.2 Retrieval baselines on all cohorts

Table 3 evaluates BM25-only, dense-only, and their RRF fusion under one harness (fresh recomputation against the released corpus snapshot) across every published cohort. Two facts emerge. First, neither single signal dominates: on the core set dense vs. BM25 is statistically indistinguishable (Wilcoxon $p = 0.257$; over all 1,092 questions $p = 0.213$), while the fusion is not (hybrid vs. dense $p = 1.8 \times 10^{-4}$, hybrid vs. BM25 $p = 7.9 \times 10^{-8}$). Second, the temporal filter remains the largest single factor everywhere: hybrid+TCR vs. hybrid gives $p = 8.5 \times 10^{-23}$ on the core and $p = 2.6 \times 10^{-66}$ over all 1,092 published questions. The fresh recomputation reproduces the frozen protocol artifact on the core set to Hit@5 0.8114 exactly; full-list MRR differs only in the third decimal (0.660 vs. 0.662) because the frozen numbers fuse full-corpus rank lists while this table fuses per-signal top-200 candidate lists — a per-question diff shows 277/281 questions at identical ranks and four questions shifted by one rank, and both snapshots are published rather than silently switched.

5.3 Effect of the gold adjudication chain

Table 4 separates two quantities that were conflated in the initial release. The *application effect* is the difference between the golds in the question file before and after persisting the pass-3 reanchor of TOP27031 (bug B1, Section 5.7): Hit@5 of the rerank list moves 0.794→0.797, MRR 0.5957→0.5964; Hit@1 of the hybrid list 0.406→0.409. The *sensitivity row* scores the same stored candidate lists against the 24 gold sets that were "recorded but never applied" (18 version-equivalent sets + 6 reanchor proposals, one of which a guardrail blocked): Hit@1 0.484 and MRR 0.621 on the rerank list. This row is published as a sensitivity analysis over gold-set width and is *not* a headline metric — every primary number in this paper is scored against applied golds.

5.4 Cited generation on the core set

Table 5 and Figure 3 present the four conditions. Without context the model almost never names the correct article (cite_acc 0.004) and abstains on 54.8% of questions. Adding fused context lifts cite_acc to 0.488; temporal filtering pushes it to **0.601** (+0.114 absolute, McNemar $p = 5.8 \times 10^{-5}$; all comparisons against no-RAG reach $p < 10^{-40}$). Rerank+TCR raises cite_acc to 0.604 ($p = 7.3 \times 10^{-4}$ vs. hybrid) and ctx_hit5 to 0.850. The ctx_hit5 values were recomputed in the integrity audit against the applied golds from the stored top-5 context lists (they previously retained run-time values recorded before the regold chain — bug B5); the recomputation (208/281 = 0.740, 229/281 = 0.815, 238/280 = 0.850) is consistent with the retrieval Hit@5 of the matching configurations (hybrid exact at 0.7402; +TCR within one question of 0.8114, the stored run-time top-5 and the fresh full-list recompute disagreeing on a single question), and the original run-time values are retained under `ctx_hit5_runtime_pre_pass_3` in the released artifact. Abstention stays near 11% whenever context is present. Disclosed weaknesses: one question (NEW90253) is excluded from the rerank+TCR condition because its candidate pool emptied after temporal filtering (n=280); stored answers are verbatim and occasionally contain CJK characters emitted by the 3B model (11/281 no-RAG, ≤5/281 in the RAG branches) — no post-hoc cleaning is applied.

5.5 Extension cohorts: rounds 1–4

Table 6 summarises all three extension cohorts. On round 1 (106 questions), TCR raises Hit@1 from 0.547 to 0.689 and Hit@5 from 0.906 to 0.925 (Wilcoxon $p = 2.1 \times 10^{-5}$); reranking alone is not significant ($p = 0.42$), and rerank+TCR at 0.651 is not either ($p = 0.71$ vs. TCR) — published as-is. On round 2 — rarer and older 1996–2015 statutes — every metric drops visibly: TCR raises Hit@5 from 0.703 to 0.781 and MRR 0.517→0.648 ($p = 4.0 \times 10^{-5}$); rerank+TCR reaches Hit@1 **0.703** ($p = 1.4 \times 10^{-3}$ vs. TCR). For generation, the same improvement chain appears (0.000→0.531 cite_acc; McNemar $p = 1.2 \times 10^{-10}$), although the span_F1 of rerank+TCR beats pure TCR with-

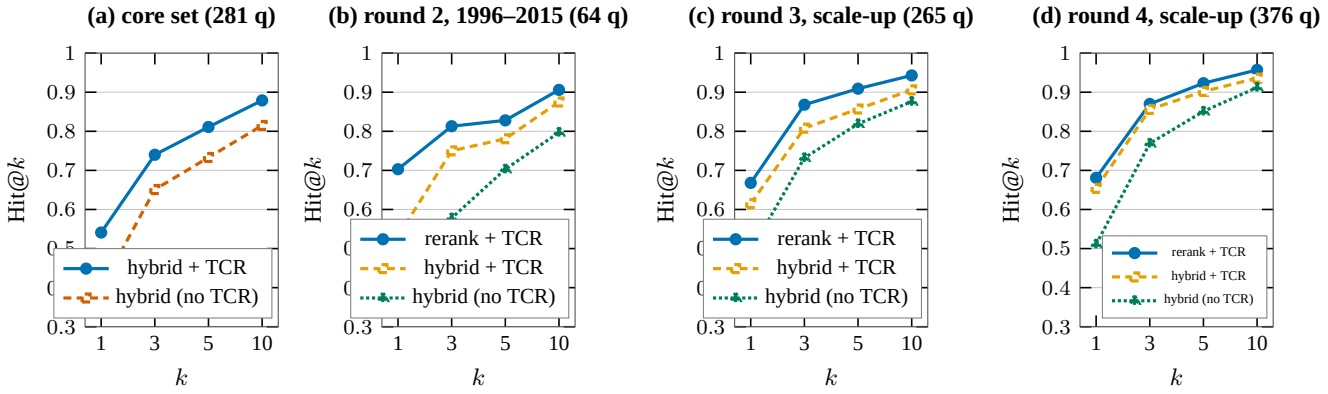


Figure 2. Hit@k curves under the shared retrieval harness: (a) the core set with and without temporal filtering; (b) the 1996–2015 cohort; (c) the round-3 and (d) the round-4 scale-up cohorts. The same stored passage vectors and RRF fusion code produce every curve.

Table 3. Retrieval baselines under one fresh-recompute harness: strict gold-passage matching, MRR within the fused candidate window; BM25 (k_1=1.2, b=0.75), dense (768-dim stored passage vectors), RRF k=60 fusion, TCR = validity-window filter.

	Core (281)			Round 1 (106)			Round 2 (64)			Round 3 (265)			Round 4 (376)			All (1,092)		
System	Hit@1	Hit@5	MRR	Hit@1	Hit@5	MRR	Hit@1	Hit@5	MRR	Hit@1	Hit@5	MRR	Hit@1	Hit@5	MRR	Hit@1	Hit@5	MRR
BM25 only	0.331	0.644	0.472	0.462	0.877	0.645	0.344	0.672	0.495	0.453	0.743	0.577	0.476	0.753	0.608	0.424	0.730	0.562
Dense only	0.356	0.666	0.495	0.453	0.840	0.614	0.359	0.766	0.519	0.472	0.793	0.606	0.460	0.787	0.611	0.430	0.761	0.575
Hybrid (RRF)	0.399	0.733	0.546	0.509	0.906	0.677	0.359	0.688	0.517	0.494	0.819	0.630	0.511	0.851	0.660	0.469	0.809	0.617
Hybrid + TCR	**0.537**	**0.811**	**0.660**	**0.660**	**0.925**	**0.775**	**0.531**	**0.781**	**0.648**	**0.615**	**0.857**	**0.724**	**0.654**	**0.902**	**0.763**	**0.608**	**0.863**	**0.721**

Table 4. Gold-set effect and sensitivity on the core set: same stored candidate lists (rerank / hybrid, 30 candidates, MRR@30), scored against different gold sets (281 questions).

	Rerank list			Hybrid list		
Gold set	Hit@1	Hit@5	MRR	Hit@1	Hit@5	MRR
Applied: before persist	0.448	0.794	0.596	0.406	0.737	0.548
Applied: after persist	0.448	**0.797**	**0.596**	**0.409**	**0.740**	**0.552**
24 recorded, never applied	0.484	0.808	0.621	0.441	0.744	0.571

Table 5. Cited generation on the core set (Qwen2.5-3B-Instruct, greedy, 180 tokens; ctx_hit5 scored against applied golds).

Condition	cite_acc	span_F1	abstain	ctx_hit5	n
No-RAG	0.004	0.325	0.548	—	281
RAG hybrid	0.488	0.466	0.110	0.740	281
RAG hybrid + TCR	**0.601**	0.490	0.107	0.815	281
RAG rerank + TCR	**0.604**	**0.490**	0.111	**0.850**	280

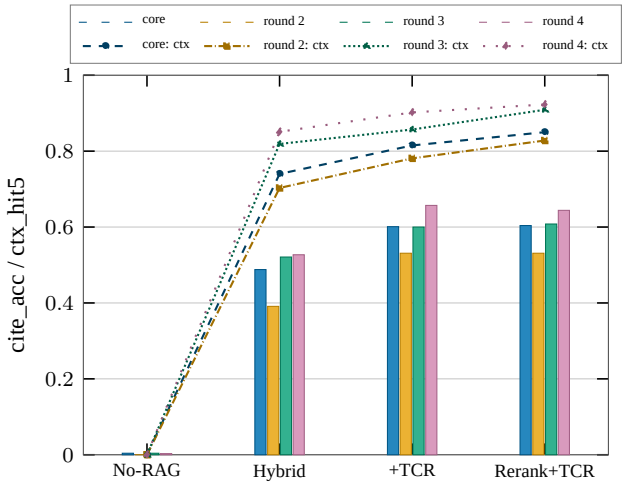


Figure 3. cite_acc (bars) and ctx_hit5 (lines) by condition on the core set (281 q) and all three extension cohorts with generation runs (rounds 2–4). Bars of one colour group belong to one cohort; the bar–line gap shows the headroom left at the generation step in every cohort.

out reaching significance (Wilcoxon $p = 0.083$). The round-2 `regolded` stratum (11 questions) is the hardest slice of the entire benchmark (Hit@1 0.091, Hit@5 0.364) — a direct consequence of anchoring golds on old statutes, labelled publicly so users can stratify. Round 3 (265 questions over 19 statutes, the scale-up cohort) behaves like a blend of both: TCR raises Hit@1 0.494→0.615 and Hit@5 0.819→0.857 ($p = 5.7\times10^{-17}$), and rerank+TCR reaches Hit@1 **0.668** and Hit@5 **0.909** ($p = 5.2\times10^{-3}$ vs. TCR, the first cohort where reranking adds a significant gain on top of TCR). Round 4 (376 questions, the largest cohort) confirms the pattern on fresh scale: TCR raises Hit@1 0.511→0.654 and Hit@5 0.851→0.902 ($p = 6.3\times10^{-22}$); rerank+TCR reaches Hit@1 **0.681** and Hit@5 **0.923**, though the gain over TCR alone does not reach significance here ($p = 0.10$, published as-is). The round-4 `regolded` stratum (13 questions) shows the same low-Hit@1 signature (0.231) as the other cohorts, and — unlike the four disclosed round-2 cases — *all* round-4 regolded anchors lie inside their validity windows at `as_of`. Round-3 cited generation, run afterwards under the identical Qwen2.5-3B-Instruct protocol (Table 7), completes the picture: cite_acc rises from 0.004 without context to 0.521 with fused context and **0.600** after temporal filtering (McNemar $p = 3.8\times10^{-3}$ vs. hybrid), and 0.608 with rerank+TCR — not significant over TCR on this cohort ($p = 0.89$), in contrast to the retrieval side where rerank+TCR was significant ($p = 5.2\times10^{-3}$). The per-configuration ctx_hit5 equals the retrieval Hit@5 exactly (0.819 / 0.857 / 0.909), reproducing the cross-pipeline consistency check of the core set, and abstention falls from 46.0% (no context) to 15–20% whenever context is present. Round 4 shows the strongest

Table 6. Extension cohorts: retrieval Hit rates and paired Wilcoxon statistics.

	R1 (n=106)		R2 (n=64)		R3 (n=265)		R4 (n=376)	
Configuration	Hit@1	Hit@5	Hit@1	Hit@5	Hit@1	Hit@5	Hit@1	Hit@5
Hybrid (base)	0.547	0.906	0.359	0.703	0.494	0.819	0.511	0.851
Hybrid + TCR	**0.689**	**0.925**	0.531	0.781	0.615	0.857	0.654	0.902
Rerank	0.585	0.906	0.516	0.797	0.555	0.879	0.588	0.875
Rerank + TCR	0.651	0.943	**0.703**	**0.828**	**0.668**	**0.909**	**0.681**	**0.923**

MRR@30 by row (base / +TCR / rerank / rerank+TCR): round 1 — 0.700 / 0.790 / 0.726 / 0.779; round 2 — 0.517 / 0.648 / 0.654 / 0.766; round 3 — 0.629 / 0.724 / 0.694 / 0.775; round 4 — 0.660 / 0.763 / 0.716 / 0.787. p (TCR vs. base): 2.1×10^{-5} (round 1), 4.0×10^{-5} (round 2), 5.7×10^{-17} (round 3), 6.3×10^{-22} (round 4); p (rerank+TCR vs. TCR): 0.71 (round 1, ns), 1.4×10^{-3} (round 2), 5.2×10^{-3} (round 3), 0.10 (round 4, ns). Hit@5 by row — round 2: 0.703 / 0.781 / 0.797 / 0.828; round 3: 0.819 / 0.857 / 0.879 / 0.909; round 4: 0.851 / 0.902 / 0.875 / 0.923.

Table 7. Cited generation on round 3 (265 questions; Qwen2.5-3B-Instruct, greedy, 180 tokens — the exact protocol of Table 5).

Condition	cite_acc	span_F1	abstain	ctx_hit5	n
No-RAG	0.004	0.338	0.460	—	265
RAG hybrid	0.521	0.444	0.196	0.819	265
RAG hybrid + TCR	**0.600**	**0.475**	0.155	0.857	265
RAG rerank + TCR	**0.608**	0.470	**0.151**	**0.909**	265

McNemar on cite_acc: +TCR vs. hybrid $p = 3.8 \times 10^{-3}$; rerank+TCR vs. TCR $p = 0.89$ (ns); no-RAG vs. hybrid+TCR $p = 5.5 \times 10^{-48}$. Wilcoxon on span_F1: +TCR vs. hybrid $p = 1.5 \times 10^{-4}$.

temporal effect of all cohorts (Table 8): cite_acc 0.003 (no context) → 0.527 (hybrid) → **0.657** after temporal filtering — a $+0.130$ absolute gain (McNemar $p = 3.2 \times 10^{-10}$) — with rerank+TCR at 0.644, statistically indistinguishable from TCR ($p = 0.635$); ctx_hit5 again equals retrieval Hit@5 exactly (0.851 / 0.902 / 0.923).

5.6 Statistical summary

Every primary comparison carries a paired test: (i) retrieval, TCR vs. base: $p = 2.1 \times 10^{-5}$ (round 1), $p = 4.0 \times 10^{-5}$ (round 2), $p = 5.7 \times 10^{-17}$ (round 3), $p = 6.3 \times 10^{-22}$ (round 4), $p = 2.6 \times 10^{-66}$ (all 1,092); (ii) rerank+TCR vs. TCR: $p = 1.4 \times 10^{-3}$ (round 2), $p = 5.2 \times 10^{-3}$ (round 3), $p = 0.10$ (round 4, ns); (iii) generation, hybrid+TCR vs. hybrid: McNemar $p = 5.8 \times 10^{-5}$ (core), $p = 0.012$ (round 2); no-RAG vs. hybrid+TCR: $p < 10^{-40}$ on both cohorts; (iv) round-3 generation, hybrid+TCR vs. hybrid: McNemar $p = 3.8 \times 10^{-3}$, rerank+TCR vs. TCR $p = 0.89$ (ns), no-RAG vs. hybrid+TCR $p = 5.5 \times 10^{-48}$; (v) round-4 generation: $p = 3.2 \times 10^{-10}$, rerank+TCR vs. TCR $p = 0.635$ (ns), no-RAG vs. hybrid+TCR $p = 1.8 \times 10^{-74}$. Non-significant comparisons are reported as such rather than omitted.

5.7 Data audit and reproduction

Before freezing the release we ran an integrity audit over all published artifacts under a *reproduce-then-patch* rule: every metric was recomputed independently from per-question artifacts, and a patch was applied only after the recomputation matched the published number to four decimal places. The audit confirmed EXT1/EXT2 exact matches, exact repro and gold-effect cells, and correct passage and cohort sizes by direct counting. Three real data defects were found and fixed

Table 8. Cited generation on round 4 (376 questions; identical protocol).

Condition	cite_acc	span_F1	abstain	ctx_hit5	n
No-RAG	0.003	0.339	0.465	—	376
RAG hybrid	0.527	0.468	0.168	0.851	376
RAG hybrid + TCR	**0.657**	**0.502**	0.144	0.902	376
RAG rerank + TCR	0.644	0.493	**0.141**	**0.923**	376

McNemar on cite_acc: +TCR vs. hybrid $p = 3.2 \times 10^{-10}$; rerank+TCR vs. TCR $p = 0.635$ (ns); no-RAG vs. hybrid+TCR $p = 1.8 \times 10^{-74}$. Wilcoxon on span_F1: +TCR vs. hybrid $p = 1.0 \times 10^{-7}$.

with logs (`CHANGES.json`, 30 changes): (B1) the pass-3 reanchor of TOP27031 was never persisted into the question file (stale gold at rank 31, correct gold at rank 1) — persisted, tier moved flagged→regolded, all core metrics recomputed on applied golds; (B2) `gold_adjudicated.json` mixed 24 recorded-but-never-applied gold sets into the applied map — separated into `proposals_not_applied`; (B3) the human adjudication kit lacked `as_of` in 50/60 rows and carried stale golds in 7/60 rows — the kit was regenerated in sync with applied golds. Additionally ctx_hit5 was recomputed (stale-value bug B5) and cross-checks exactly against per-configuration retrieval Hit@5. After the patch: core Hit@5 0.808→0.811, MRR 0.659→0.662, TCR cite_acc unchanged at 0.601; EXT1/EXT2 unchanged. The full process, old→new numbers, and the new SHA-256 hashes are published in `AUDIT_2026-11-01.md` — we argue audits of this kind should be mandatory for every public benchmark before submission.

6 Discussion and Limitations

Data limitations. (1) Narrow domain: all questions belong to civil and civil-procedure law; 20 documents and 1,092 questions are small next to multi-domain benchmarks. (2) Non-uniform gold tiers: 26 core questions are flagged and excluded from primary metrics; the regolded stratum (22 questions) has systematically low Hit@1 — users must stratify, as we do. (3) The independent judge agreement reaches only 0.571 on the 60-question sample; this is a model-vs-model check on an 8-candidate selection task — a task with many defensible choices — and *does not replace* human evaluation; the 60-question human kit (with guidelines and Cohen/Fleiss κ) is released and is the required next step before claiming a human-validated benchmark. (4) Questions are LLM-generated and inherit the generator's biases; mitigated by vector deduplication, two-tier auditing, and public quarantine, but not eliminated. (5) Four round-2 `regolded` anchors (EXT20028, EXT20039, EXT20113, EXT20123 — 4 of the 11) are temporally out of scope at the question's `as_of`: three of them (EXT20028, EXT20039, EXT20113) point to consolidated texts whose validity window starts *after* `as_of`, while EXT20123 points to a BLDS 1995 passage whose validity window *ended* on 2006-01-01, six years before its `as_of`. Each anchor was accepted by the adjudication chain as the legal basis of the underlying question, but all four fall outside the validity windows at the stated dates, so TCR filtering penalises them by construction. We disclose them rather than silently re-anchoring a second time; under TCR they con-

tribute to the low regolded-stratum numbers reported above.

Evaluation limitations. The experiments use one 3B generator and one reranker; no larger LLM baseline or domain fine-tuning is included. TCR relies on a manually verified temporal graph; a wrong validity window propagates directly into the filter. Three MRR semantics (full-list / MRR@10 / MRR@30) are annotated separately but not yet unified — normalising to full-list is the next technical task. cite_acc measures naming the correct *article numbers*, not quoting the correct *passage*.

Ethics and intended use. The corpus consists of public legal texts with per-passage provenance and source-type labels; no personal data is involved. The benchmark is intended for evaluating retrieval and QA systems on Vietnamese statutes; answers produced by models evaluated on it are not legal advice.

7 Conclusion

ViCivil-TKG demonstrates that two principles — temporal-validity checking and multi-stage gold adjudication with public quarantine — operate consistently on Vietnamese legal data and both yield measurable gains: TCR improves cite_acc by +0.114 points ($p < 10^{-4}$) and retrieval Hit@5 on every cohort; the adjudication chain eliminates the unverifiable-gold state entirely and makes gold-anchor quality measurable per stratum. The third methodological contribution is the reproduce-then-patch audit procedure with a full public change log. The 1,092-question benchmark with 1,029 quarantined questions, full provenance, a human adjudication kit, and a GPU-free reproduction script is released for community scrutiny and extension; next steps in priority order: human adjudication on the stratified sample (kit available), unifying MRR semantics, extending to circular-level documents and adjacent legal domains, and benchmarking larger generators.

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